

Lung Cancer Dataset Analysis

Key Findings:

- Smoking is strongly associated with lung disease.
- Pollution exposure significantly increases disease risk.
- Older individuals are more likely to develop lung disease.
- Logistic regression identified Smoking, Pollution, and Age as significant predictors.
- AUC ≈ 0.787 indicating good model performance.

Important Statistical Results:

- Chi-square test p-value < 0.001 for Smoking vs Lung Disease
- Odds Ratio for Smoking ≈ 5.03
- Accuracy $\approx 71.2\%$
- Sensitivity $\approx 75.8\%$
- Specificity $\approx 66.1\%$

Public Health Implications:

- Smoking cessation programs should be strengthened.
- Pollution control policies are necessary.
- Early screening is important for smokers and older adults.